

**A Case for Adopting Machine Learning and Deep Learning Technologies in Kenya's
Healthcare System for Brain Tumor Detection.**



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Dissertation

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Declaration

I declare that all the content contained in this research is original and has never been submitted to any other institution for the award of a degree or diploma. Moreover, references are aptly applied to other authors whose works were pivotal in this study. This body of work aligns with Dublin Business School's academic honesty policy.

Acknowledgment

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Abstract

Brain tumors pose a dire health risk in Sub-Saharan Africa, particularly in Kenya. This research aims to develop and optimize models for detecting brain tumors in MRI images, thus minimizing diagnostic delays. The study employed a Cross Industry Standard Process for Data Mining (CRISP-DM) methodology. It utilized a dataset of 3,260 MRI images categorized into four classes: meningioma tumors, pituitary tumors, glioma tumors, and images with no tumors.

Different machine learning and deep learning models, including Logistic Regression, Support Vector Machine, Decision Tree, a custom Convolutional Neural Network, and pre-trained models such as InceptionV3, VGG19, and Xception, were explored. The performance metrics revealed that deep learning models significantly outperformed machine learning models.

The findings in this research underscore the need for deep learning techniques to enhance the accuracy of brain tumor classification. Hence, this elucidates their potential to reduce delays in diagnosis and improve patient outcomes. The results obtained are thus expected to contribute to the body of knowledge in early and accurate diagnosis of brain tumors in Kenya and the global medical community.

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Chapter 1

1.0 Introduction

Sub-Saharan Africa faces numerous socio-economic challenges, from political instability in Sudan to rising inflation across the region. Inadequate medical infrastructure and services exacerbate these issues, leading to underreported diseases and an incomplete understanding of the region's health crisis. Despite limited data, cancer mortality rates in Sub-Saharan Africa have increased by over 40% since 2000, while many developed countries have seen declines (Olivier et al., 2023). Brain cancer, commonly referred to as brain tumors, has significantly contributed to this increase, particularly in Kenya, which comprises a youthful demographic. It is concerning that Dr. Muriithi points out that it is this demographic that is most affected by brain tumors (Dr. Muriithi, 2009).

A brain tumor is characterized by abnormal cell growth, which develops into a mass in the brain. Brain tumors are either benign (non-cancerous) or malignant (cancerous). When brain tumors develop from the brain itself, they are regarded as primary brain tumors. If their origin is attributed to growth from other parts of the body then they are secondary or metastatic brain tumors (Abdusalomov, et al., 2023). In clinical practice, pituitary, glioma, and meningioma tumors are the most common primary brain tumors detected in patients (Nodirov, et al., 2022). Notably, gliomas constitute the most prevalent malignant tumors.

Despite the advancements in brain tumor diagnosis, such as the development of Magnetic Resonance Imaging, it remains a challenge. This is demonstrated by the fact that MRI scans are often subjected to manual analysis by radiologists and other medical experts to detect and segment brain tumors. This inadvertently exposes them to personal interpretation, introducing bias in the analysis, which may be erroneous. In addition, manual analysis is often time-consuming, leading to delayed diagnosis. Therefore, a more robust system of diagnosis is necessary for early and accurate diagnosis of brain tumors.

Recently, this has become possible through leveraging novel technologies in Artificial Intelligence. Machine learning and deep learning have become the fundamental pillars upon which brain tumor detection systems are continually being developed and implemented. It is

through these technologies that neurology experts can make more informed diagnoses and prognoses for patients. This research delves into how automated brain tumor detection systems can be developed through machine learning and deep learning to alleviate the burden on existing neurological facilities and reduce brain tumor-related fatalities in Kenya. Therefore, a crisp understanding of what they are is essential towards the development of this research.

1.1 Aim of Study

This research aims to investigate the potential of machine learning and deep learning technologies in developing and optimizing automated brain tumor detection systems. This would improve diagnostic accuracy and decrease fatalities from brain tumors in Kenya.

1.2 Objectives of the Study

- I. To explore different machine learning and deep learning models for brain tumor detection and identify the optimal model.
- II. Evaluate the efficacy of the built models in detecting brain tumors using MRI scans.

1.3 Problem Statement

Over the last decade, public fundraising has become increasingly common in Kenya to help family members and friends settle medical bills for brain tumor treatments. This reflects the socio-economic challenges faced by patients and their families, which have been exacerbated by rising brain tumor cases and subsequent increases in mortality rates, as highlighted by a 2022 report from the Global Cancer Observatory.

According to Wanyoike (2004) and John et al. (2021), delayed diagnosis of brain tumors is a salient cause of brain tumor-related deaths. The delays are primarily due to inadequate diagnostic tools and the high costs associated with diagnosis.

1.4 Research Question

How can machine learning and deep learning models be optimized to improve the accuracy and efficiency of brain tumor diagnosis, leading to a reduction in diagnostic delays and associated mortality rates in Kenya?

Chapter 2

2.0 Literature review

2.1 Introduction

Thanks to automated approaches, early detection of brain tumors is becoming increasingly possible. Modern-day researchers are continually devising new techniques to create these automated systems. This section, therefore, delves into previous works carried out by other researchers and presents their research findings.

2.2 Transfer Learning Approach

One recent approach by Abdusalomov et al. (2023) leverages transfer learning to build upon existing knowledge. They fine-tuned the pre-trained YOLOv7 model, a real-time object detection system, for brain tumor identification in MRI scans. They incorporated a Convolution Block Attention Module (CBAM) into the model to improve feature representation and object class discrimination. This module enhances the model's performance by capturing spatial and channel dimensions from the input data. Moreover, the module enables the model to focus on relevant information and suppress image noise, allowing better generalization and reducing image variation. YOLOv7 was suitable for their model development due to its proven efficacy in detecting brain tumors.

The researchers used a dataset of 10,288 MRI scans from a public source. Data augmentation techniques were implemented during preprocessing to increase the sample size artificially. Additionally, a Gaussian blur filter was applied to reduce image noise.

Abdusalomov et al.'s network architecture consists of two key components: a backbone network and a head network. The backbone network handles feature extraction. It comprises convolutional layers, batch normalization, Max Pooling, and E-ELAN modules, which perform convolution and batching, apply SiLU activation, and facilitate modular computation for feature addition, respectively. The head network, containing E-ELAN units and a Feature Pyramid Network Architecture, handles feature extraction and fusion.

Precision, recall, sensitivity, specificity, accuracy, and F1-score metrics were used to test the model's performance. The percentage scores of each are 99.5%, 99.3%, 99.3%, 99.4%, 99.5%,

and 99.4%, respectively, clearly demonstrating Abdusalomov et al.'s model's effectiveness in brain tumor detection.

Owida et al. delve into this research area and introduce a novel model architecture derived from the EfficientNetB3, a pre-trained Convolutional Neural Network (2024). The authors use 7,023 MRI images obtained from public databases to implement this model. The images are resized to 244 x 244 pixels during preprocessing to ensure consistent image dimensions and simplify their processing and subsequent analysis. The preprocessed images were augmented using the ImageDataGenerator from Keras utility to enhance the model's ability to generalize and handle variations. Additionally, segmentation – the division of an image into various regions of interest – was employed during this phase.

To tailor the EfficientNetB3 model for brain tumor classification, the researchers fine-tuned it by adding batch normalization layers, a regularized dense layer with 0.01 weights, and a dropout regularization with a 0.5 dropout rate. The introduction of the batch normalization layers standardized the input data, thereby increasing the speed and stability of model training. In contrast, the use of a regularized dense layer prevented overfitting by penalizing weights and activities in the model. Nonetheless, incorporating the dropout technique significantly reduced the model's reliance on specific features, increasing its generalization.

The evaluation metrics used demonstrate this model's efficiency in brain tumor classification, with a training and validation accuracy of 100%. Thus, the model learned the key features and patterns in the data, which enabled it to accurately classify brain tumors into glioma, meningioma, and pituitary and establish the absence of tumors in an MRI image.

Indra et al. (2023) similarly explore the efficacy of transfer learning in detecting brain tumors. They conduct their experiments using EfficientNetB0, MobileNetV2, and Xception pre-trained CNN architectures. To develop the models, a dataset consisting of 2,880 MRI images categorized into meningioma tumors, pituitary tumors, glioma tumors, and normal brain cells is utilized. Resolution rescaling and data augmentation techniques are employed during preprocessing to improve image quality and increase the sample size to improve model training, validation, and testing.

Before model development, the researchers modified the mentioned CNN architectures by adding an additional layer of the Pooling2D layer, introducing dropout rates, and a ReLU activation function to enhance model performance. Recall, accuracy, and precision were used as the performance metrics to compare the results. Their findings affirm that the models are suitable for deployment into automated systems by achieving 97% accuracy in identifying the type of tumor. The MobileNetV2, in this case, achieved 98.8% accuracy, while the EfficientNetB0 and Xception garnered 98.33% and 97.36% accuracy (Indra et al., 2023).

Jaiswal et al. demonstrate further research into transfer learning, where multiple neural network architectures are leveraged to identify the most apt model to detect brain tumors (2023). The ResNet 152, Xception, VGG 16, Inception V3, and MobileNet V2 constitute this study's proposed transfer learning models. As is evident in the previously mentioned studies, this study's focal point is the development of models that can distinguish three brain tumors: meningioma, glioma, and pituitary from normal brain tissue. Similarly, the dataset applied comprises 3,064 MRI images.

To smoothen and remove noise from the images, a Gaussian filter is used, after which they are converted into grayscale and passed through the Contrast Limited Adaptive Histogram Equalization (CLAHE) algorithm to adjust their contrast (Jaiswal et al., 2023) during image preprocessing. Moreover, contour drawing is applied as an image-enhancing technique to improve the edge features of the image, which are pivotal to enhanced convergence.

Following data preprocessing, the outlined pre-trained neural network architectures are used for classification by fine-tuning either their linear or fully connected layers. The evaluation parameters in this study are accuracy, precision, recall, and F1 score. From their subsequent results, after model training and testing, the Xception model is deemed the most optimal brain detection system as it records an accuracy of 98.04%. The ResNet 152, Inception V3, VGG 16, and MobileNet V2 achieve accuracy rates of 97.35%, 93.46%, 75.82%, and 36.6% respectively.

2.3 Convolutional Neural Network Approach

Contrastingly, Pereira et al. propose a different alternative to segment brain tumors: low-grade glioma and high-grade glioma tumors (2016). Low-grade glioma tumors are less fatal than higher-grade glioma tumors. Their technique is anchored on convolution neural networks. Small convolutional kernels (3×3) in the convolutional layers characterize the uniqueness of their technique. The small kernels allow for the stacking of more convolutional layers, thus enabling the establishment of more complex patterns in the MRI scans.

Furthermore, this approach effectively captures local features as the kernel design subjects it to a smaller receptive field. Also, the model employs an intensity normalization method to remove the bias field distortion in the images. Data augmentation was performed during preprocessing, and the images were specifically rotated across multiples of 90 degrees to avoid the distribution of a wrong class to a patch. The results of the various experiments conducted using this model showed a 4.6% mean gain by introducing intensity normalization and conclusively established that LReLU was more effective than ReLU during the training of the CNN model.

Similarly, the study by Mahmud et al. explores using Convolutional Neural Networks to build automated brain tumor detection systems (2023). This study compares the CNN model performance against the Resnet-50, VGG16, and Inception V3 models. The training, validation, and testing data comprising of 3,264 MRI images is sourced from a public database. The CNN model is designed to classify glioma tumors, meningioma tumors, pituitary tumors, and the absence of tumors in the MRI images. Data augmentation was conducted during preprocessing. Laplacian and Gaussian filters were implemented to ensure quality input was fed into the model to remove noise from the MRI images.

The model received input images with a size of 32×32 pixels. The first two convolutional layers used 16 and 32 filters with a 3×3 kernel to capture low-level features from the MRI images. A max-pooling layer was added following each convolutional layer to reduce the spatial dimensions of the data. The fully connected dense layer used a SoftMax activation function to

predict the class probabilities. Their model also implemented a dropout with a 0.5 rate after each convolutional layer, apart from the last one, to prevent overfitting.

To analyze the CNN model's performance, accuracy, recall, and area under the curve were selected as the performance evaluation metrics. Compared to the aforementioned models, this model ranked first across all the metrics, achieving 93.3%, 91.13%, and 98.43%, respectively.

As evidenced in the experiments performed by Musallam et al., their proposed Deep Convolutional Neural Network (DCNN), outperformed CNN models based on VGG16, VGG19, and a hybrid CNN-SVM model (2022). The DCNN achieved a 98.2% accuracy in detecting brain tumors, while the other models recorded an accuracy of 95.4%, 95.4%, and 90.7%, respectively. The high accuracy can be attributed to their 3-step preprocessing approach and the model's architecture, designed from scratch.

The first step involves removing texts and black regions by cropping 100 pixels around the MRI images. Secondly, they employ a non-local mean algorithm to remove image noise, as it is a better image enhancer than the Gaussian blur filter and Median filter (Musallam et al., 2021). Lastly, histogram equalization is utilized to improve image contrast, consequently allowing to detect minute details.

The designed DCNN model has two parts: the convolutional and classifier parts. The former has ten convolutional layers, five batch normalization layers, and four max-pooling layers, whereas the latter comprises three dense layers and two dropout layers.

Another notable study by Almadhoun and Naser demonstrates the development of a CNN model from inception to its testing to detect brain tumors (2022). The dataset employed in this study had 10,000 MRI images categorized into 2: presence of brain tumor and absence of brain tumor. The created model comprised 12 convolutional layers and a fully connected hidden layer (Almadhoun & Naser, 2022). A softmax activation was used in the output layer to determine the probability of each class, and Adam optimization was utilized to optimize the network. Notably, a checkpoint was added to the model to save the highest validation accuracy, as training over a certain number of epochs may lead to overfitting (Almadhoun & Naser, 2022).

After training, the model was tested on 1,000 MRI images not used during training and achieved an accuracy of 98%. Its performance was thereafter compared to VGG 16, ResNet50, MobileNet, and Inception V3. Following this comparison, it was determined that the best-performing models were the VGG 16 and Inception V3, which achieved 99% accuracy. ResNet50 and MobileNet had 98% and 88% accuracy, respectively.

2.4 Machine Learning Approach

Support Vector Machines (SVM) have emerged as a powerful tool in the field of medical image classification, particularly for brain tumor diagnosis using MRI scans. Recent studies, as highlighted in the works of Gurusamy and Subramaniam (2017) have proved the effectiveness of SVM classifiers in accurately distinguishing tumor-related abnormalities from healthy brain tissue. Their study focused on binary classification as normal brain tissue was to be distinguished from brain tumor MRI images. Moreover, they designed an SVM that was structured as a feed forward network.

Their study performed a comparative analysis, and SVM showed superior performance. It achieved an accuracy of 98% in predicting positive values and 98% in predicting negative values. From their findings, this classifier clearly offers a robust framework for brain tumor detection.

Other than SVM, decision tree models are considered robust for classification tasks. Naik and Patel (2013) evidence this. To assess their effectiveness in identifying brain tumors, the researchers conducted experiments using binary classification on CT scans. These experiments aimed to distinguish between normal scans (absence of tumors) and abnormal scans indicative of tumor presence. This study used the Naive Bayes and the Decision Tree classifier. A confusion matrix was leveraged as the primary evaluation tool. As such, precision, recall, specificity, and accuracy metrics identified the optimal classifier. The decision tree classifier achieved 96% accuracy, 93% sensitivity, and 100% precision, outperforming the Naive Bayes classifier.

A table showing their respective accuracy is presented below to compare the models reviewed comprehensively. Furthermore, through the insights drawn from this literature review, this study had a multifaceted approach to enhance brain tumor detection. The review has proven CNNs and transfer learning models as high-performing models in detecting brain tumors.

Consequently, the multifaceted approach encompasses machine learning models such as SVM and deep learning models, including CNNs and transfer learning models. Moreover, through data preprocessing, data augmentation, and model fine-tuning, the study sought to enhance the accuracy and generalization ability of the classification models that were developed. Through this approach, the study leveraged the strengths of the different methodologies to build robust brain tumor detection systems.

Table 1: A Comparison of Model Performance Reviewed in this Section

Machine Learning Approach		
Author	Model	Accuracy
Gurusamy & Subramaniam (2017)	SVM	98%
Naik and Patel (2013)	Decision Tree	98%
Transfer Learning Approach		
Abdusalomov et al. (2023)	YOLOv7	99.50%
Owida et al. (2024)	EfficientNetB3	100%
Indra et al. (2023)	EfficientNetB0, MobileNetV2, Xception	98.33%, 98.8% & 97.36%
Jaiswal et al. (2023)	Resnet 152, Xception, VGG16, InceptionV3, MobileNetV2	97.35%, 98.04%, 93.46%, 75.82% & 36.6%
Convolutional Neural Network (Custom Architecture)		
Mahmud et al. (2023)	CNN	93.30%
Musallam et al. (2022)	DCNN	98.20%
Almadhoun & Naser (2022)	CNN	98%
Pereira et al. (2016)	<i>Segmentation Technique</i>	-

Chapter 3

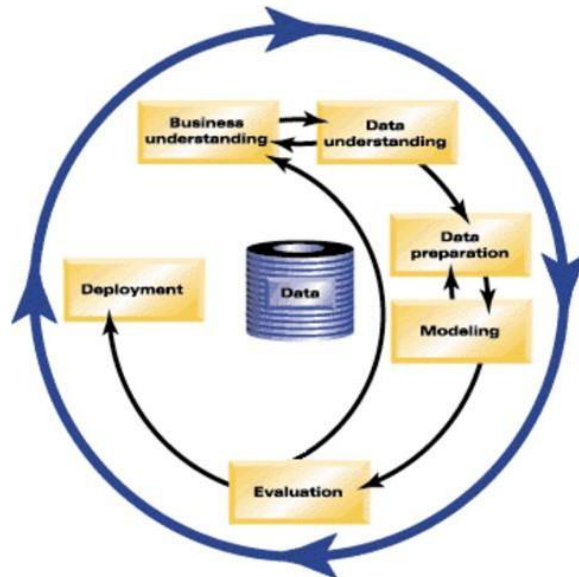
3.0 Research Methodology

3.1 Introduction

A robust methodology must be defined for any scientific research to be considered coherent. The applicable methodology offers a framework that allows research to be well organized and reliable and to provide an avenue to build the body of knowledge in a particular field. Consequently, this research employs a Cross Industry Standard Process for Data Mining (CRISP-DM) methodology, the standard approach in data analytics.

3.2 CRISP-DM Methodology

The adoption of this methodology is attributed to its reliability when conducting data mining projects. It is also considered a cost-effective strategy and easy to iterate. The process is categorized into six main phases, shown in the figure below and subsequently discussed, that form the crux of this investigation.



Source: <https://www.kaggle.com/discussions/getting-started/404902>

3.2.1 Phase 1: Business Understanding

This research addresses the critical need for accurate and automated brain tumor classification within the medical field. Early and precise diagnosis is crucial for effective treatment planning and improved patient outcomes. By leveraging deep learning techniques, this research

developed models capable of automatically classifying different types of brain tumors, which can aid medical professionals during diagnosis.

3.2.2 Phase 2: Data Understanding

The dataset employed in this project comprises brain MRI images classified into four distinct tumor types: glioma tumor, meningioma tumor, no tumor, and pituitary tumor. A bar chart visualization, illustrated in the figure below, was generated to assess class distribution, revealing an imbalance among the classes. Furthermore, the analysis demonstrated significant variability in image sizes within the dataset. Most images, 2,309, had a resolution of 512 x 512 pixels. However, a smaller subset of images displayed different resolutions, with 46 images measuring 236 x 236 pixels and 30 images measuring 225 x 225 pixels.



Figure 1: Training image distribution

3.2.3 Phase 3: Data Preparation

This stage incorporates all tasks involved in developing the final dataset for model development. These tasks include data tabulation, recording, attribute selection, cleaning, and transformation. Notably, the tasks performed in this stage can be repeated and executed without a predefined order.

3.2.4 Phase 4: Modelling

Different algorithms are selected and implemented during modeling to manipulate the defined dataset. Actions such as parameter tuning to obtain optimal values are performed here.

3.2.5 Phase 5: Evaluation

The preferred models are developed, followed by the ascertainment of their performance in achieving the primary objectives of the analysis.

3.2.6 Phase 6: Deployment

This research does not implement the deployment phase, as the primary focus is model development and evaluation. Deployment in a real-world healthcare setting would require further considerations and resources, including integrating the model into existing clinical infrastructure, ensuring data privacy and security, and acquiring necessary regulatory approvals.

3.3 Research Design

Research design serves as a blueprint for any research investigation, as it describes the framework for collecting and analyzing data to actualize the proposed research objectives. The choice of a research design impacts the quality and generalizability of research findings. Researchers can employ an exploratory, cross-sectional, or experimental design from the array of accepted research designs. A study's aim and primary objectives form the tenets guiding the choice of a research design.

3.3.1 Type of Research Design in this Study

According to Zubair, the active manipulation of independent variables is the hallmark feature of an experimental study (2023). Therefore, as this research involved manipulating independent variables to make predictions of dependent variables, an experimental approach was deemed

appropriate. In this case, the independent variables are the image features learned during model training and then used to classify either the type of brain tumor or the absence thereof.

3.3.2 Research Approach

Furthermore, the design implementation was based on a mixed-method approach as qualitative and quantitative aspects of data were applied. The former describes descriptive data, which could be in the form of words, images, and sounds, while the latter describes numeric data. In their inherent form, MRI scans are images categorized as qualitative. However, images are constituted of quantitative pixels. They are the main features extracted by the deep learning models to make qualitative predictions, in this case, describing the nature of the brain tumor.

3.4 Data Collection Methods

To conduct this study, secondary data in the form of MRI scans from a public dataset, provided in this link <https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri>, "Brain Tumor Classification (MRI)" by Sartaj Bhuvaji, was utilized. The dataset consists of 3,260 MRI images. The MRI images are categorized into four different classes that depict the type of brain tumors: glioma tumors, meningioma tumors, pituitary tumors, and images with no tumors. Additionally, the MRI images are split into training and testing datasets; thus, it simplifies the model validation process. The selection of this dataset is informed by the high quality and quantity of the images, which are sufficient to perform the subsequent model experiments.

It is also noteworthy that the data was cleaned and ready for model development. The MRI images were also physically verified by a doctor, as indicated in the description of this dataset, therefore making them suitable for this research.

3.4.1 Data Sampling

Data sampling is pivotal to ensure apt model generalization in seen and unseen data during data preparation for model training and testing. As the dataset was already split, it was shuffled to prevent overfitting during training. Data augmentation techniques were applied to increase the sample size and enhance generalization. This included rotation, contrast adjustment, and random zooming of the images. A validation dataset was also introduced to allow monitoring of

the model performance during model training. Therefore, validation data was extracted from the training dataset in an 80:20 split using the train and test split module.

3.5 Data Analysis Methods

Although all phases of research are crucial, data analysis is one of the cornerstones of any research. Regarding this research, data analysis encompassed the processes that extracted meaning from the MRI image dataset. This consisted of data preprocessing, exploring and visualizing the data to uncover inherent patterns, and applying machine and deep learning algorithms to classify the MRI images into the respective brain tumors present or otherwise absent. Thus, combining these data analysis processes enabled the formulation of a concrete framework for developing a brain tumor classification system.

3.5.1 Techniques and Tools

This study used machine and deep learning algorithms. The application of these algorithms formed the basis of the criterion used to identify the most appropriate brain tumor classification system. The machine learning algorithms under consideration were the Support Vector Machine, logistic regression, and decision tree classifier. In addition, convolutional neural network algorithms were used to implement deep learning, and transfer learning models were also leveraged.

3.5.2 Data Visualization Tools

Matplotlib proved to be a pivotal tool in this study in creating detailed plots and visualization of class imbalances, data distribution, and data evaluation metrics.

3.5.3 Software

Python was chosen as the primary programming language for this study due to its rich ecosystem of libraries for machine and deep learning, as well as data visualization. The key libraries that were utilized included Pandas, NumPy, Scikit-Learn, OpenCV (cv2), Matplotlib, TensorFlow, and Keras. Recognizing the computationally intensive data analysis task, Google Colab was leveraged as a cloud-based resource, specifically utilizing its powerful GPUs. The L4 runtime, with its 22.5 GB GPU RAM, 53.0 GB system RAM, and 122.6 GB disk space, provided a

robust environment for seamless model training by significantly accelerating the training process.

3.6 Data Analysis Steps

3.6.1 Data Preprocessing

The cv2 library was used to import MRI images. A custom load function was developed to streamline the image-loading process from the respective training and testing directories. This function efficiently organized the loaded images into a list, subsequently converted into a NumPy array for efficient data manipulation during model development. Each image was then matched with its corresponding label. The following figure showcases a sample of the loaded images.

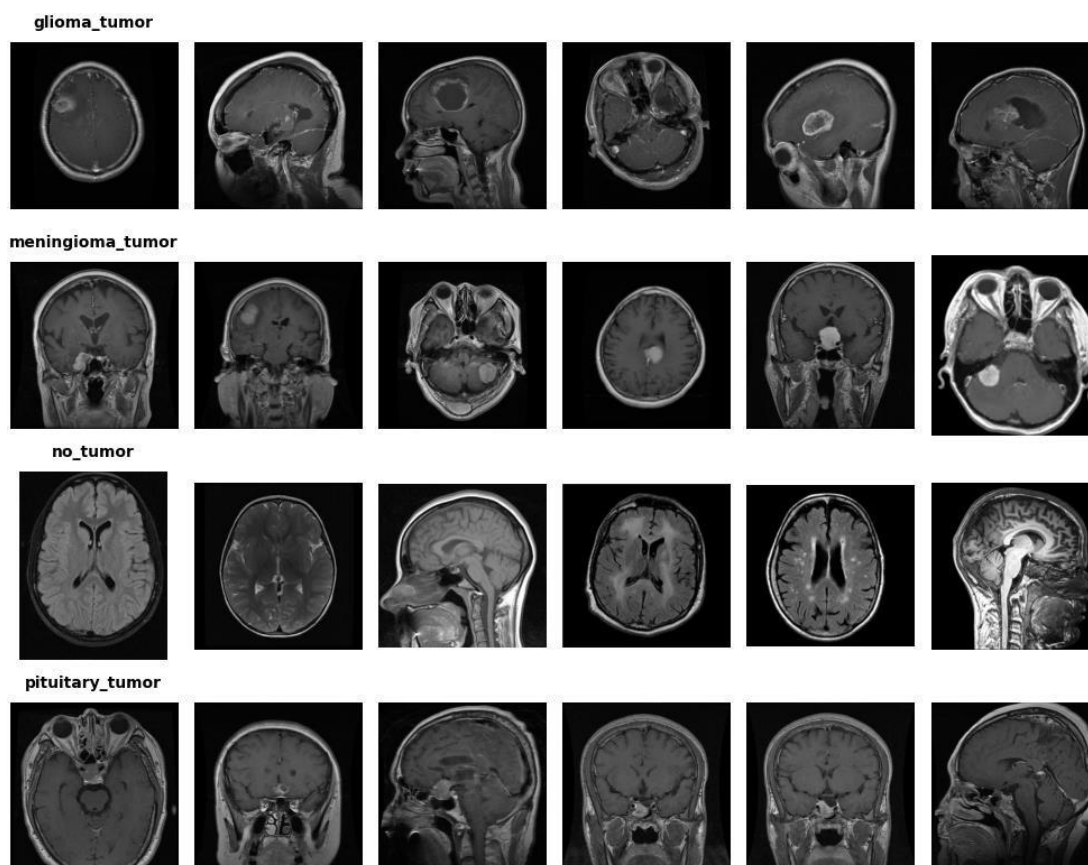


Figure 2: Loading training images

Moreover, to prepare the images for model training, the following preprocessing steps were undertaken:

i. Image Resizing

Since the images were not consistent, they were resized to 256 x 256 pixels to ensure uniformity, as neural networks require the inputs to be similar in size and thus learn patterns effectively.

ii. Image Cropping

Since tumors are concentrated within a particular region, isolating the region and removing background information is pivotal (this can be considered noise removal). A cropping function was thus used to isolate the region of interest within the images.

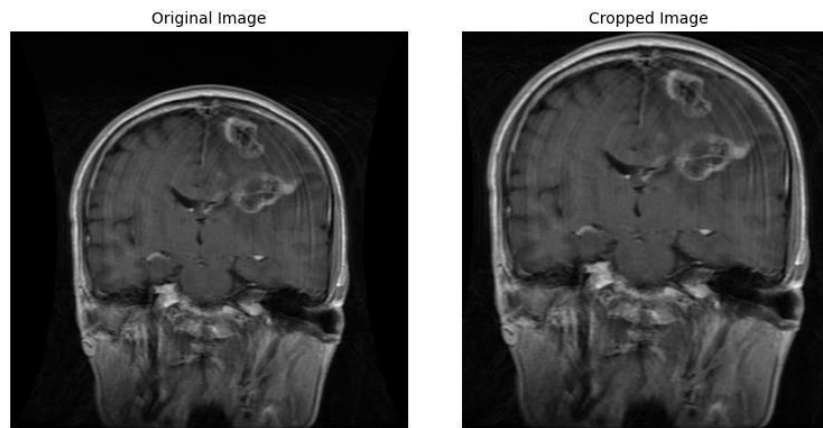


Figure 3: Cropped MRI image

iii. Data Augmentation

The performance of machine learning models is intricately linked to the quality and quantity of the training data. Models trained on limited data or imbalanced datasets are susceptible to overfitting and struggle to generalize effectively. To address these limitations, data augmentation was employed. As noted by Hernandez García (2020), data augmentation is crucial for enhancing model performance. Moreover, Owida et al. (2024) and Abdusalomov et al. (2023) apply data augmentation in the experiments outlined in the literature review.

This technique generated modified copies of existing data, thereby expanding the training dataset. It increased the sample size hence enabled models to generalize better. This project used the ImageDataGenerator API in TensorFlow to perform data augmentation. This technique introduced variations in the training data through rotations, shifts, flips, and brightness adjustments, facilitating the development of more robust and generalizable models.

In addition to the above preprocessing steps, the image labels were one hot encoded to enable the correct interpretation of the target labels by the model during model development and prediction. The MRI images were then split into training and validation data using a train-and-test split in an 80:20 ratio. The latter would allow the evaluation of the model performance during training.

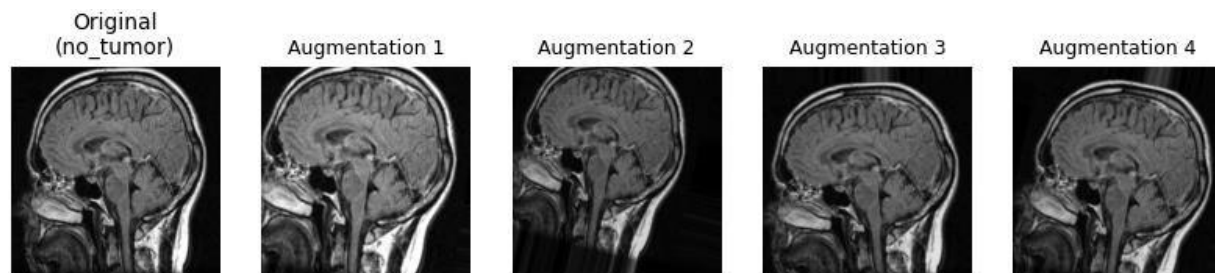


Figure 4: Image showing data augmentation

3.6.2 Feature Extraction and Selection

In this study, feature extraction is inherently performed by the neural network architectures. The convolutional layers of the models automatically learn relevant features from the input images, thus eliminating the need for manual feature engineering. Feature selection is also implicitly handled during training as the neural networks distinguish which features are most discriminative for classification.

3.6.3 Model Development

This project explores various modeling techniques, encompassing deep learning and traditional machine learning approaches, to effectively classify brain tumors from MRI images. The selection of diverse models allows for a comparative analysis and identification of the most suitable approach for this specific task.

Deep Learning Models

Convolutional Neural Network (CNN)

A custom feedforward CNN architecture forms the foundation of this project's deep learning approach. This model leverages the inherent ability of CNNs to learn spatial hierarchies of features from images, making them well-suited for tumor classification. The architecture of a CNN comprises of:

- **Input Layer:** This layer feeds the model an input comprising the images' pixels. The layer defines the shape of an image's pixel: the width, height, and channels. In this study, the images have a width and height of 256 x 256 and 3 (RGB) channels.
- **Convolutional Layer:** This layer is at the core of CNN as it is responsible for convolution. In this layer, the feature representations of an image are learned. It comprises filters, which are crucial in convolution as they extract features from the images. Furthermore, the layer consists of kernels, 3 x 3, that are applied in the computation of feature maps, which are convolution outputs. The kernels in the first convolutional layer are designed to detect low-level features. Conversely, in the deeper layers, the kernels encode more abstract features of the inputs.

Since kernels lead to information loss around an input's borders, 'same' padding is used in the convolution layer to enlarge the input by adding a value, 0, around it (Li et al., 2021). A low stride is also used to capture distinct input features.

Non-linear activations are also integral in the convolutional layer as they enable the detection of non-linear features in the network. As pointed out by Gu et al., the ReLu activation has been empirically observed to perform better than other functions like sigmoid and tanh (2017). Hence, it is applied in this study's CNN.

- **Pooling Layer:** Similarly, the pooling layer is an indispensable component of the CNN. It reduces the spatial dimensions of the output feature maps after convolution and decreases parameter count and data volume. Therefore, the computational complexity and model training time are decreased. A singular pooling layer is used in each convolutional block, as

stacking multiple layers would have a blurring effect that would invariably affect model performance (Zhao & Zhang, 2024).

Moreover, max pooling with a 2×2 kernel and a stride of 2 is utilized, as using a greater kernel size would decrease the model performance (O'Shea & Nash, 2015). In addition, it is a widely used method in many image processing tasks (Graham, 2015) and is also apt at capturing local features.

- **Dropout Layer:** Overfitting is a common problem in neural networks. Consequently, the dropout regularization technique is leveraged to address it. The layer randomly dropped units(neurons) from the neural network during model training to prevent co-adaptation of the neurons. The utilization of this layer is further demonstrated by Srivastava et al., who postulate that dropout improves the performance of neural networks in vision classification tasks (2014).
- **Global Average Pooling Layer:** The Global Average Pooling layer acts as a dimensionality reduction step. It averages the feature values across the feature maps' spatial dimensions (width and height), resulting in a single value for each feature map. This effectively summarizes the spatial information while preserving the most important features.
- **Dense Layer (Fully Connected Layer):** In this layer, all neurons are fully connected to the neurons of the preceding layer. This dense connectivity allows it to capture complex relationships between the high-level features extracted by the preceding layers. In this study's model, the number of neurons in the fully connected layer is defined in the subsequent chapter.
- **Output Layer:** The output layer is the final layer in the model. It is responsible for producing the final classification of an MRI image. Since this is a multiclass classifier, a SoftMax function is utilized for its appropriateness in multi-classification tasks. More succinctly, it generates the probability score of each class, and the one with the highest score is given as the model's prediction.

Transfer Learning with Pre-trained Models

Pre-trained CNNs, namely InceptionV3, Xception, and VGG19, were used in transfer learning to investigate optimal deep learning models further. The TensorFlow transfer learning framework was applied to define their architecture.

- I. **VGG19:** This CNN model is anchored on the idea that increasing the depth of a neural network significantly increases its final performance. Moreover, as it uses small kernels (Li et al., 2021), it is appropriate for model development in this research.
- II. **Inception V3:** The inception V3 network is part of GoogleNet's neural network architectures built upon the advancements of inceptionV2. The model's efficacy is improved by introducing factorized convolutions that significantly accelerate model training and allow creation of deeper neural networks. This architectural improvement, coupled with its demonstrated high accuracy - as evidenced by Almadhoum & Nasser, who achieved 99% predictive accuracy (2022)- makes Inception V3 a suitable candidate for model development in this study.
- III. **Xception:** Xception is a neural network that utilizes depthwise separable convolutions, processing input channels simultaneously. As Chollet (2017) highlighted, this approach significantly reduces computational cost. Xception's high performance, achieving 98% accuracy in Jaiswal et al. experiments (2023), makes it an ideal choice for model development in this study.

Model Training in the Deep Learning Models

The following parameters are defined when training the custom CNN and the transfer learning models,

- **Optimizer:** Optimizers adjust a model's weights and biases during training. Their primary role is to minimize the loss function, which quantifies the difference between the model's predictions and the true MRI images' labels. By effectively reducing the loss function, optimizers enhance the models' prediction.

Adam Optimizer was used during model training as it is suitable for high-dimensional data, and its implementation is straightforward (Kingma & Ba, 2017).

- **Learning Rate:** The learning rate controls the step size taken by the optimizer during each iteration. Lower learning rates are widely used in deep learning to ensure stable training and prevent model divergence. Furthermore, smaller learning rates improve a model's generalization accuracy (Wilson & Martinez, 2001). Thus, a learning rate of 0.001 is defined for model training in the custom CNN. The transfer learning rates are 0.0001, as this is the default in the TensorFlow transfer learning APIs.
- **Loss Function:** Categorical Cross-entropy loss is applied in these models as it is suited for multiclass classification tasks. This loss function measures the difference between the outcomes' predicted and actual probabilities (labels). Since accurate diagnosis of brain tumors is crucial, this loss function is preferred in this study. It strongly reinforces correct classifications by penalizing misclassifications (Terven et al., 2024). Its emphasis on minimizing prediction errors is thus vital for ensuring the reliability of these brain tumor classifiers if successfully deployed in clinical settings.

Traditional Machine Learning Models

In addition to deep learning, traditional machine learning models were considered to provide a comparative perspective and empirical evidence for why deep learning models are preferable in image classification tasks. The selection of the logistic regression model was arbitrary as the researcher applied his prior knowledge of machine learning. The models were tested over scaled (min-max and standard scaling), unscaled, and augmented data.

- i. **Decision Tree:** A tree-based model that makes decisions based on a series of hierarchical splits pegged on feature values. In this experiment, the model begins with a root node, the training dataset, and then applies recursive partitioning to split the data into smaller subsets (Naik & Patel, 2013). Each internal node subsequently makes a decision based on a particular feature like pixel intensity and classifies the tumor at the leaf node.
- ii. **Support Vector Machine:** As earlier discussed, SVMs are robust models that seek the optimal hyperplane to separate data points into different classes. They are effective for MRI

image classification as they are effective in high-dimensional spaces and are adept at handling complex relationships between features as characterized in MRI scans. Additionally, this model is identified for testing as it exhibits self-learning, enabling it to adapt and enhance its accuracy (Gurusamy & Subramaniam, 2017)

- iii. **Logistic Regression:** A linear model employed for binary or multiclass classification. This study's logistic regression models the relationship between image features and the probability of belonging to a particular tumor class. Similarly, the use of this model in this research is supported by the works of Kowshika and Shoban et al. (2022), who applied logistic regression to detect brain tumors and strokes, achieving a 98% accuracy.

3.6.4 Model Evaluation and Validation

Evaluating these models' efficacy in detecting brain tumors is imperative to reflect real-world scenarios. The testing dataset was used to evaluate the data while accuracy, precision, recall, and F1-score were the evaluation metrics that informed the optimal model's performance. Furthermore, a confusion matrix was utilized to understand predictive performance comprehensively. The confusion matrix visually highlighted the classification performance by summarizing the number of correct and incorrect predictions made by the models. The mentioned evaluation metrics are extracted from it and are discussed below.

Accuracy

The accuracy metric measures how many correct brain tumor predictions a model makes in a dataset. The following formula describes it:

(TP = True positives, TN = True negatives, FP = False positives, FN =False negatives)

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall/Sensitivity

Recall describes the number of actual positive cases of brain tumors detected by the model. In this study, it is a primary evaluation metric, as a high recall value decreases the occurrence of false negatives—missing a brain tumor. Furthermore, a high recall can identify tumors in the early stages, thus potentially reducing brain tumor diagnosis time.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Precision

The precision metric describes the proportion of correctly predicted positive cases of brain tumors among all images predicted as having brain tumors for a particular category of brain tumor.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{TN})$$

F1-Score

The F1 Score is a harmonic mean between precision and recall. In this study, it was used to compare the models' performance at a glance.

$$\text{F1 - Score} = 1 / ((1/\text{Recall}) + (1/\text{Precision}))$$

Following the generation of the confusion matrix, classification reports were created to summarize the evaluation metrics. The classification report is easy to interpret and analyze, making it an invaluable tool for model evaluation.

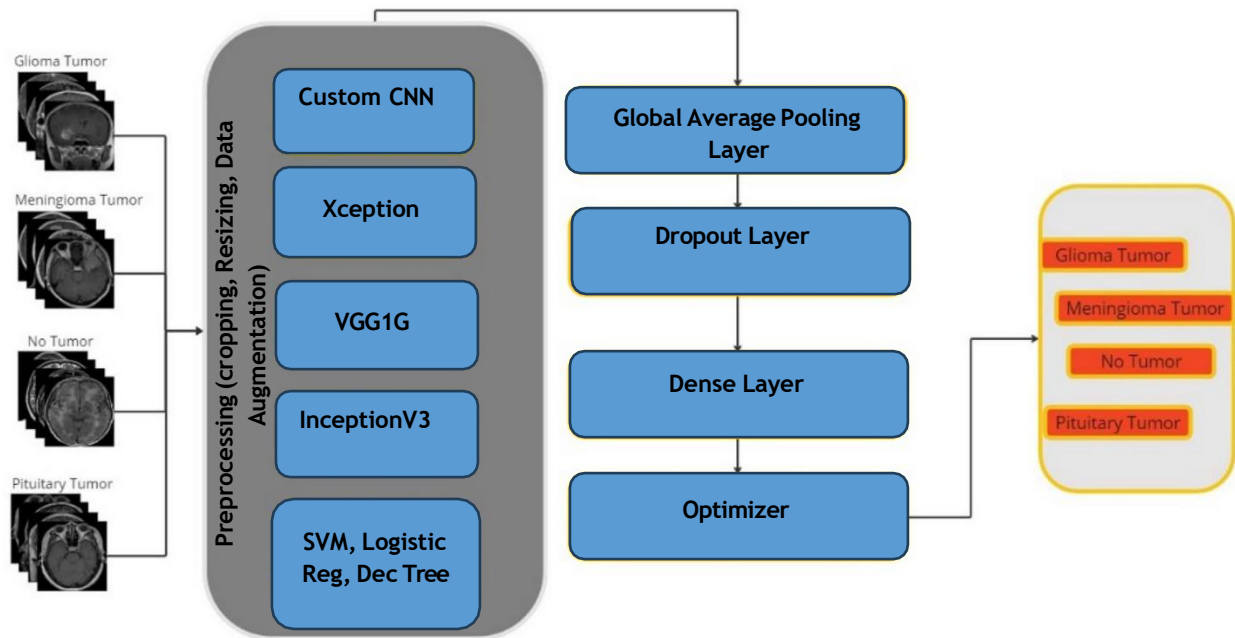


Figure 5: Diagrammatic representation of the methodology

3.7 Research Challenges

Despite using Colab GPUs during model training, in some instances, the models crashed, and thus, experimentation had to begin afresh. This was particularly the case when performing GridSearch CV hyperparameter tuning in the machine learning models. Consequently, the tuning was not performed to determine the best parameters for the machine learning models.

3.8 Conclusion

This chapter outlines the research methodology that was used in this study to classify brain tumors. The CRISP-DM ensured a well-structured, iterative, and robust process. The critical phases in the process are well-defined. Moreover, an experimental research design was executed as various experiments were conducted. In combination with the mixed-method approach, the research design enhanced the analysis process due to the quantitative and qualitative nature of the images.

Training and testing data was acquired from a secondary source obtained from Kaggle, which is publicly accessible. It comprises 3,260 images categorized into 4 categories: meningioma tumors, pituitary tumors, glioma tumors, and images without tumors.

A range of machine learning and deep learning models were explored, including Logistic Regression, SVM, Decision Tree, a custom CNN, and pre-trained models such as InceptionV3, VGG19, and Xception. Model training and evaluation were conducted using Python with libraries such as TensorFlow/Keras, NumPy, Pandas, and OpenCV. Google Colab was utilized for efficient model training and analysis.

The performance of each model was subsequently evaluated using metrics such as accuracy, precision, recall, and F1-score, with the confusion matrix providing a detailed visual representation of model predictions.

Through this methodology, the study explored machine and deep learning technologies in brain tumor detection, optimized developed models, and demonstrated the significance of these technologies in reducing delays in detecting brain tumors. This methodological framework provided the foundation to gain valuable insights in this brain tumor classification task.

3.9 Timeline and Resource Requirements

The major resource employed was cloud GPUs, whose acquisition cost the researcher incurred throughout the experiments. The research was conducted in four main phases, each with specific tasks and an estimated duration. The project took 12 weeks to complete.

Phase 1: Planning and further literature review

Phase 2: Data preparation and preprocessing

Phase 3: Model Development and Evaluation

Phase 4: Final Documentation and Submission

Chapter 4

4.0 Artefact Design & Development

4.1 Introduction

4.2 Custom Convolutional Neural Network

As earlier noted, a custom convolutional neural network is defined and used for model training.

The following is a colab extract of the model summary showing its architecture.

Model: "sequential"		
Layer (type)	Output Shape	Param #
lambda (Lambda)	(None, 256, 256, 3)	0
conv2d (Conv2D)	(None, 254, 254, 16)	448
max_pooling2d (MaxPooling2D)	(None, 127, 127, 16)	0
dropout (Dropout)	(None, 127, 127, 16)	0
conv2d_1 (Conv2D)	(None, 125, 125, 32)	4,640
max_pooling2d_1 (MaxPooling2D)	(None, 62, 62, 32)	0
dropout_1 (Dropout)	(None, 62, 62, 32)	0
conv2d_2 (Conv2D)	(None, 60, 60, 64)	18,496
max_pooling2d_2 (MaxPooling2D)	(None, 30, 30, 64)	0
dropout_2 (Dropout)	(None, 30, 30, 64)	0
conv2d_3 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_3 (MaxPooling2D)	(None, 14, 14, 128)	0
dropout_3 (Dropout)	(None, 14, 14, 128)	0
conv2d_4 (Conv2D)	(None, 12, 12, 256)	295,168
max_pooling2d_4 (MaxPooling2D)	(None, 6, 6, 256)	0
dropout_4 (Dropout)	(None, 6, 6, 256)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 256)	0
dense (Dense)	(None, 512)	131,584
dropout_5 (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 4)	2,052
Total params: 526,244 (2.01 MB)		
Trainable params: 526,244 (2.01 MB)		
Non-trainable params: 0 (0.00 B)		

Figure 6: Model Summary of CNN architecture

As shown above by the model summary table, the architecture of the CNN is defined by an initial input layer where an MRI image, with the shape (256 x 256, 3) is rescaled. The input image is rescaled to the -1, 1 range, and the batch size is defined later during model compilation. MaxPooling2D layers with a stride of 2 were integrated into the architecture to reduce the spatial dimension in the data. The model also reduces dimensions step by step through the global average pooling layer, and the dense layers are defined to classify the brain tumors into four categories. After the model definition, the model is experimented with through a series of epochs: 60, 80, and 200.

4.2.1 CNN Model Implementation

- **Input Layer:** In this layer, MRI images with a pixel size of 256 x 256 and RGB channels are fed into the model.
- **Data preprocessing:** A rescaling layer is added to the model to improve model training by rescaling the MRI images to the [-1, 1] range.
- **Convolutional layers:** To effectively extract the features from the scans, convolutional filters with increasing depth- 16, 32, 64, 128, 256- have been defined in the model. Five convolutional layers were applied in this network. Padding was defined as 'same' to ensure that output feature maps had the same dimensions. Moreover, the ReLu activation was used in this layer to introduce non-linearity in the model. It is important to emphasize that the reasons that informed the selection of this activation function are pointed out in the previous chapter.
- **Max Pooling Layers:** Max pooling layers with a 2 x 2 pooling window were used in pooling of the MRI scans. This layer as highlighted reduces the spatial dimensions of the output feature maps thus increasing computational efficiency. Moreover, the layer was added to the architecture as it focuses on crucial activations and therefore improving model learning.
- **Dropout Layer:** Additionally, dropouts have applied for regularization to prevent overfitting of the model. Dropout rates were determined by running a Keras hyperparameter tuner.
- **Dense Layer:** The first dense layer, which has 512 neurons, adds another layer of non-linearity and allows the model to learn complex patterns between the features extracted by

the convolutional layers. The ReLu activation applied in this layer is responsible for this non-linearity.

- **Output layer:** The final dense layer has four neurons, one for each class. SoftMax activation is utilized as this is a multiclass classification task. It outputs the probabilities of the four classes during prediction.

4.3 Transfer Learning

As previously outlined, transfer learning leverages preexisting knowledge to make predictions on a new dataset. This study involves two distinct phases: training and fine-tuning using three pre-trained models: VGG19, InceptionV3, and Xception. These models are pre-trained on the ImageNet weights.

The training phase commences with loading and feature extraction from the training dataset. Individual brain MRI images are loaded, and their pixel matrices are converted into vector representations, effectively extracting the image features.

Preprocessing and data augmentation techniques are then applied to enhance data variation, which improves model generalization. These techniques include cropping to isolate the region of interest, random rotation, translation, zoom, and contrast adjustments. Subsequently, all images are resized to a uniform size of 256x256 pixels to ensure compatibility with the model architecture.

The convolutional base of the pre-trained model is initially frozen to prevent drastic changes during the initial training stages. The extracted image features are then passed through a custom classification head. This head incorporates global average pooling to reduce dimensionality, a dropout layer to mitigate overfitting, and a dense layer with SoftMax activation to predict the probabilities of each tumor class: glioma, meningioma, no tumor, and pituitary. The model is trained using the labeled training data and the categorical cross-entropy loss function.

After initial training, the pre-trained models' convolutional base is partially unfrozen to enable fine-tuning. This fine-tuning step allows the model to further adapt its feature extraction

capabilities to the specific characteristics of the brain tumor dataset, leading to improved performance as will be observed in the training and validation curves.

A new brain MRI image is presented to the trained model in the testing phase. The same feature extraction process applied during training converts the image into a vector representation. This feature vector is then fed into the trained model. The model processes the features and assigns a probability score to each brain tumor class. Finally, the class with the highest probability score is selected as the predicted brain tumor class for the input image.

This two-phase process effectively leverages transfer learning to build a robust and accurate brain tumor classification model.

4.3.1 VGG19 Architecture & Implementation

VGG19 is a renowned convolutional neural network architecture characterized by its substantial depth, comprising 19 weight layers. Developed by the Visual Geometry Group at Oxford University, it achieved cutting-edge results in image recognition on the ImageNet dataset in 2014.

The model consists of a series of 3x3 convolutional filters that capture fine-grained details while preserving spatial information. This structure is further enhanced by a repetitive pattern of convolutional layers followed by max-pooling layers, resulting in a clear and easily comprehensible design. Applying the ReLU activation functions introduces non-linearity, allowing the network to learn complex patterns from the data.

Model Implementation

- **Input Layer:** The input into the model consists of MRI images with a shape of 256x256 pixels and three color channels.
- **Preprocessing:** Images are preprocessed by normalizing and rescaling pixel values to match the expected input range of the pre-trained VGG19 model $[-1, 1]$.
- **VGG19 Base Model (Feature Extraction):** The preprocessed images are fed into the frozen VGG19 base model. The pre-trained model's ability to extract high-level features learned

from the ImageNet dataset is leveraged in this step. The base model's weights are frozen to prevent them from being updated during initial training.

- **Global Average Pooling:** The output feature maps from the VGG19 base model are passed through a Global Average Pooling layer where spatial information across the feature maps is summarized. Moreover, dimensionality is reduced as pertinent spatial relationships are maintained.
- **Dropout:** A Dropout layer is introduced to randomly deactivate a fraction of neurons during training. This regularization technique helps prevent overfitting, improving the model's generalization ability to unseen data.
- **Dense Layer:** A Dense layer with four output neurons corresponding to the four brain tumor classes (glioma, meningioma, no tumor, pituitary) is added. This layer uses a SoftMax activation function to produce a probability distribution across the classes, representing the model's confidence in each prediction.
- **Output Layer:** The model outputs a probability distribution over the four brain tumor classes. The class with the highest probability is the predicted class for the input image.

Model Architecture

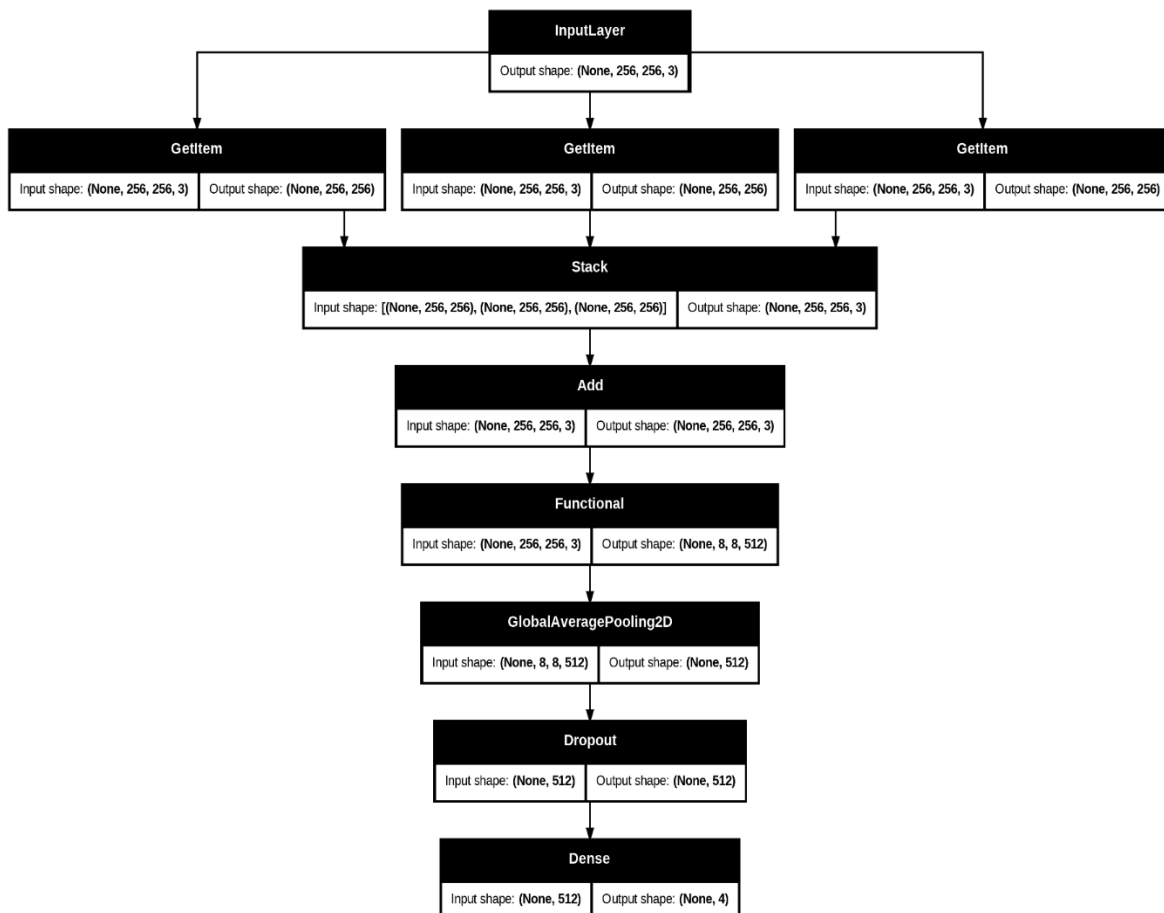


Figure 7: VGG19 Model Architecture

4.3.2 Xception Architecture and Implementation

The Xception architecture, an evolution of Inception models, leverages depthwise separable convolutions for improved efficiency. Unlike traditional convolutions that process all input channels simultaneously, depthwise separable convolutions first convolve on each channel independently, followed by a pointwise convolution to combine the results. This approach reduces the computational costs and enhances feature extraction by isolating and emphasizing unique patterns within each channel.

Model Implementation

- **Input Layer:** The model receives input brain MRI images with a shape of 256x256 pixels and three color channels.
- **Preprocessing:** Images are preprocessed. This step involves normalizing and rescaling pixel values to match the expected input range of the pre-trained model, which is usually $[-1, 1]$.
- **Xception Base Model:** The preprocessed images are then fed into the frozen Xception base model. In this layer, the pre-trained model's ability to extract high-level features learned from the ImageNet dataset is harnessed to extract the input images' features. Freezing the base model's weights prevents them from being updated during initial training, ensuring stability and efficient knowledge transfer.
- **Global Average Pooling:** The output feature maps from the Xception base model are passed through a Global Average Pooling layer. This operation summarizes spatial information across the feature maps, reducing dimensionality while retaining important spatial relationships.
- **Dropout:** A dropout layer is introduced randomly to deactivate a fraction of neurons during training. This regularization technique helps prevent overfitting and improves the model's generalization ability to unseen data.
- **Dense Layer:** A Dense layer with four output neurons corresponding to the four brain tumor classes is added. This layer uses a SoftMax activation function to produce a probability distribution across the classes, representing the model's confidence in each prediction.

- **Output:** The model outputs a probability distribution for each input brain MRI image, indicating the likelihood of it belonging to each of the four tumor classes. The class with the highest probability is selected as the predicted class for the input image.

Model Architecture

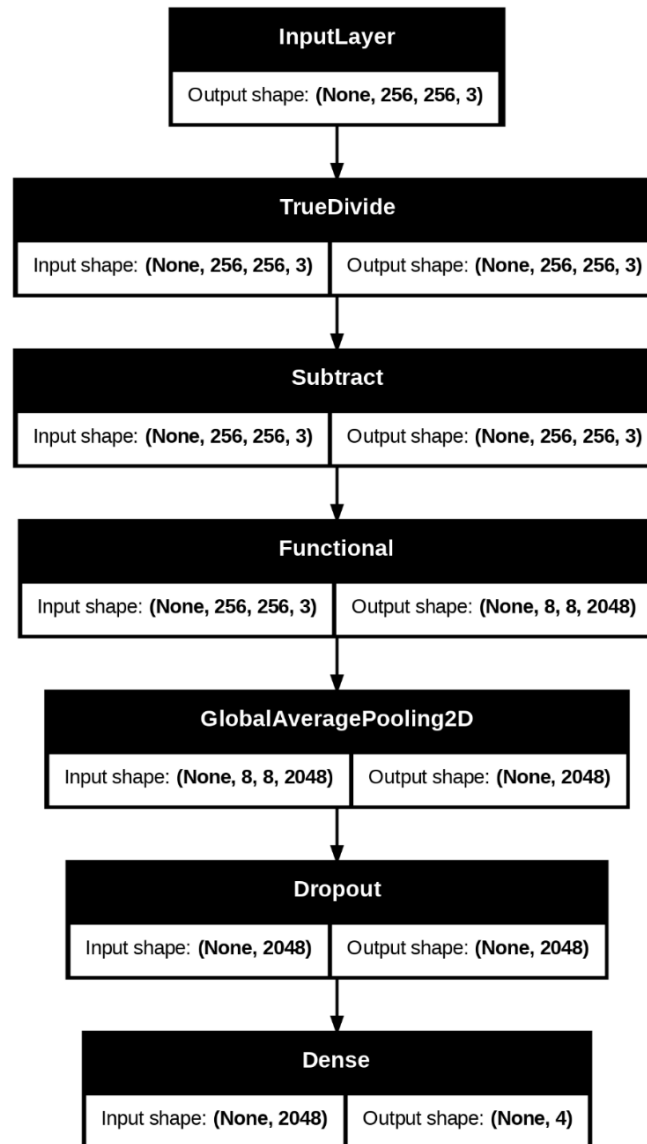


Figure 8: Xception model architecture

4.3.3 InceptionV3 Architecture and Implementation

The InceptionV3 architecture is a deep convolutional neural network that efficiently utilizes computational resources and effectively captures multi-scale features. Its Inception module allows for the simultaneous execution of multiple convolution and pooling operations. Thus, the model learns input features at varying levels of detail. By combining parallel processing with advanced techniques for filter factorization and dimensionality reduction, InceptionV3 achieves high accuracy without incurring excessive computational costs.

Model Implementation

- **Input Stage:** MRI images are fed into the model as input. These images are 256x256 pixels and contain three color channels (RGB).
- **Image Preparation:** The images are prepared for model manipulation during preprocessing. This step utilizes the `keras.applications.inception_v3.preprocess_input` function to adjust the pixel values to align with the expected input range of the pre-trained InceptionV3 model `[-1, 1]`.
- **InceptionV3 Base:** The preprocessed images are then passed to the core of the model, the InceptionV3 base. This pre-trained component serves as a feature extractor. Therefore, it leverages its prior knowledge from the ImageNet dataset to identify high-level features within the brain MRI images. The weights of the InceptionV3 base are initially frozen to maintain stability and ensure efficient knowledge transfer. Moreover, it prevents them from being modified during the early training phases.
- **Global Average Pooling Layer:** The output of the InceptionV3 base consists of feature maps, which are subsequently processed in this layer to condense their spatial dimensions and hence reduce their dimensionality.
- **Dropout Layer:** This architecture integrates a dropout layer to enhance the model's generalization to unseen data and reduce overfitting.
- **Dense Layer:** Finally, a Dense layer acts as the classification head of the model. It comprises four output neurons, each corresponding to one of the four brain tumor classes. The softmax activation function ensures that the output is a probability distribution.

Model Architecture

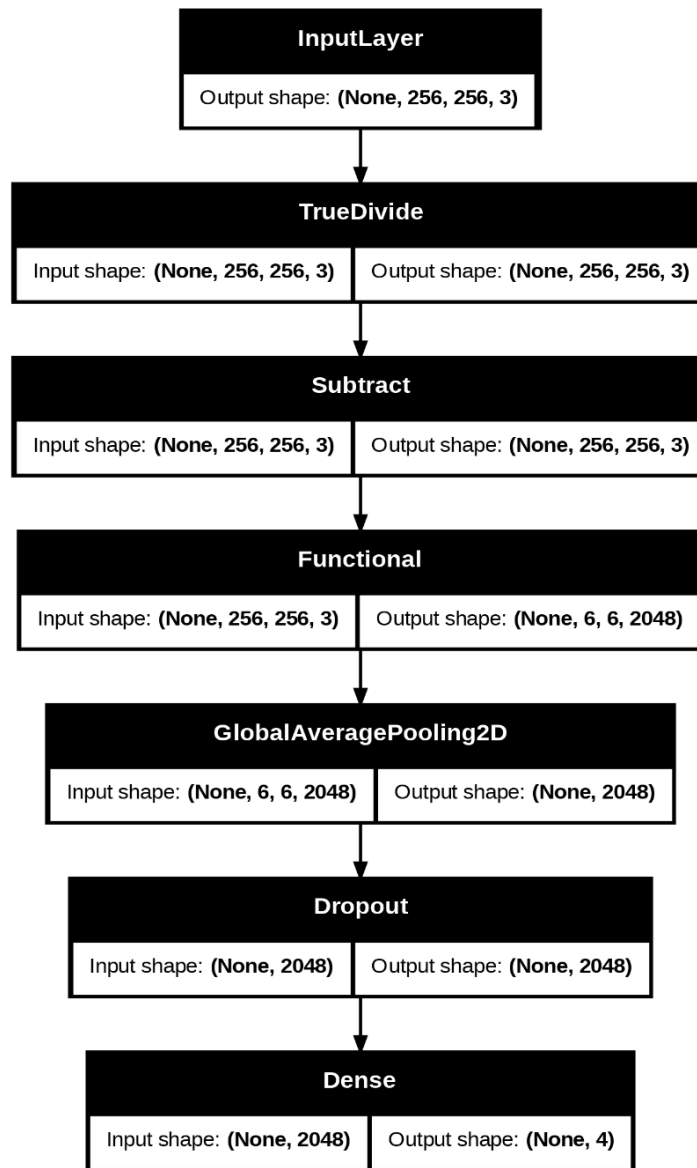


Figure 9: Inceptionv3 Architecture

Chapter 5

5.0 Model Evaluation & Results

5.1 Introduction

This chapter delves into the performance metrics of the models during experimentation. A comparative analysis of their performance determines the most optimal model. The primary tools employed for model evaluation are confusion matrix analysis and classification reports. The classification report is essential as it processes the raw data from the confusion matrix into precision, recall, F1-score, and accuracy metrics.

Moreover, given the class imbalance in the test dataset, the weighted average in the classification report was used to identify the models' precision, recall, and F1-score classification metrics. This approach mimics real-life medical scenarios requiring accurate predictions of frequently occurring tumors. The macro average would otherwise give the same importance to each class.

5.2 Confusion Matrix Analysis

The confusion matrix was indispensable throughout this study. The visualization of the models' predictions provided a detailed understanding of their performance. This critical analysis informed model tuning efforts which subsequently led to performance improvements. This section details, visually, the performance of each model.

5.2.3 Machine Learning Models

As observed in the following figures, the tested machine learning models produced the confusion matrices as illustrated. The matrices reflect the models' training under unscaled data, which was optimal compared to training with scaled and augmented data.

Logistic Regression Classifier

Although this classifier achieved 70% accuracy, the logistic regression classifier demonstrated promising results. However, like the deep learning models tested in this study, it exhibited challenges in distinguishing glioma tumors. Conversely, it was able to effectively identify positive instances of meningioma tumors and the absence of tumors in the MRI scans. The model gained commendable metrics with a precision of 71%, recall of 70%, and an F1-score of 67%.

Confusion Matrix:

```
[[ 33  25  34   8]
 [   5 101   6   3]
 [   1   2 102   0]
 [   8  15  11  40]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.70	0.33	0.45	100
1	0.71	0.88	0.78	115
2	0.67	0.97	0.79	105
3	0.78	0.54	0.64	74
accuracy			0.70	394
macro avg	0.71	0.68	0.67	394
weighted avg	0.71	0.70	0.67	394

Figure 10: Confusion matrix & Classification Report for Logistic Regression

Support Vector Machine Classifier

The SVM classifier achieves a low accuracy of 61% compared to the logistic regression classifier. The confusion matrix shows that it similarly struggled to classify the glioma tumor class, with only 21 out of 100 correctly classified. Nonetheless, due to its performance, it was not optimal for determining brain tumors.

Confusion Matrix:

```
[[ 21  39  35   5]
 [   2  95  14   4]
 [   2   1 102   0]
 [   5  34  13  22]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.70	0.21	0.32	100
1	0.56	0.83	0.67	115
2	0.62	0.97	0.76	105
3	0.71	0.30	0.42	74
accuracy			0.61	394
macro avg	0.65	0.58	0.54	394
weighted avg	0.64	0.61	0.56	394

Figure 11: Confusion matrix & Classification Report for SVM

Decision Tree Classifier

The decision tree classifier performed better than the SVM. Accurate predictions were made for the meningioma and no tumor classes, while poor performance was evident in the glioma and pituitary tumor classes. An accuracy of 65% was achieved, whereas recall was 65%.

Confusion Matrix:

```
[[ 18  31  41  10]
 [   0 106   7   2]
 [   1   0 104   0]
 [   9  20  17  28]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.64	0.18	0.28	100
1	0.68	0.92	0.78	115
2	0.62	0.99	0.76	105
3	0.70	0.38	0.49	74
accuracy			0.65	394
macro avg	0.66	0.62	0.58	394
weighted avg	0.66	0.65	0.59	394

Figure 12: Confusion matrix & Classification Report for Decision Tree Classifier

Inception V3

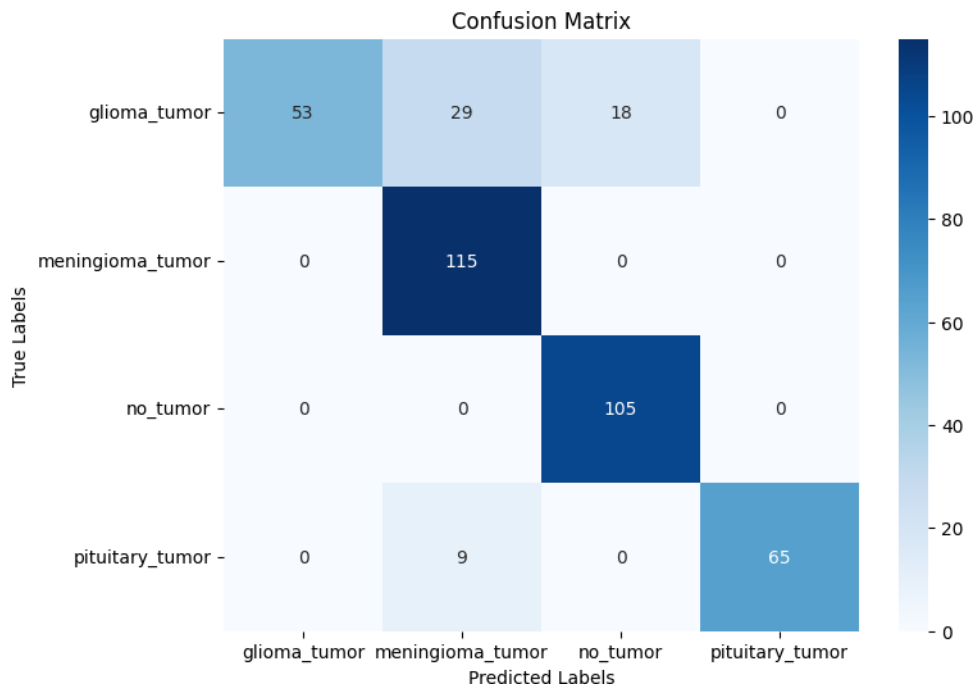


Figure 13: Confusion matrix for InceptionV3

From the confusion matrix, the meningioma tumor and no tumor classes show the highest accuracy, as the model correctly predicted all test samples, 115 and 105, respectively. Furthermore, the model accurately predicted the pituitary tumor classes, as 65 were correctly classified and only 9 were wrongly classified. However, out of the 100 glioma tumor samples, only 53 were correctly classified. It is thus observable that the model struggles to distinguish this class from the others.

Classification Report:				
	precision	recall	f1-score	support
glioma_tumor	1.00	0.53	0.69	100
meningioma_tumor	0.75	1.00	0.86	115
no_tumor	0.85	1.00	0.92	105
pituitary_tumor	1.00	0.88	0.94	74
accuracy			0.86	394
macro avg	0.90	0.85	0.85	394
weighted avg	0.89	0.86	0.85	394

Figure 14: Classification Report for InceptionV3

The InceptionV3 model demonstrates strong performance, achieving an overall accuracy of 86% and a precision of 89%, indicating a high proportion of accurate positive predictions. The model's recall stands at 86%, suggesting its effectiveness in identifying a substantial portion of the actual positive cases. An F1-score of 85% reflects a good balance between precision and recall.

VGG19

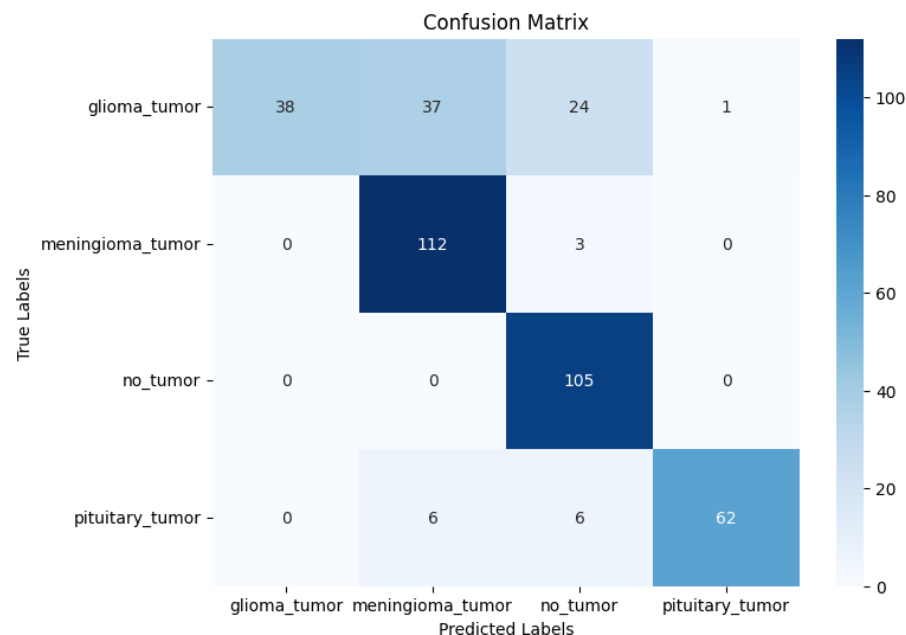
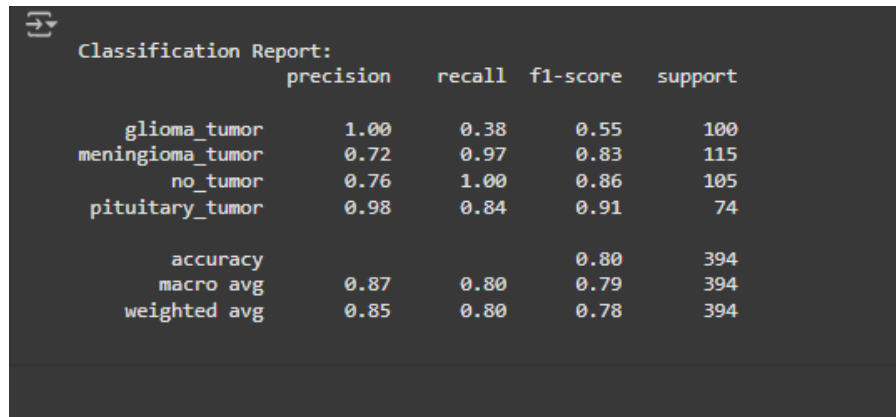


Figure 15: Confusion Matrix for VGG19

The VGG19's confusion matrix depicts a model that correctly classifies a preponderance of the instances in the right classes. The no-tumor class exemplifies the strongest prediction accuracy, where all instances are correctly classified. Out of the 105 meningioma instances, only three are wrongly classified in the no-tumor class. On the other hand, the model misclassifies six pituitary tumor instances as meningioma and six as glioma tumors.



```
Classification Report:
              precision    recall  f1-score   support

 glioma_tumor      1.00      0.38      0.55       100
 meningioma_tumor   0.72      0.97      0.83       115
      no_tumor      0.76      1.00      0.86       105
 pituitary_tumor    0.98      0.84      0.91        74

 accuracy          0.80          0.80          0.80       394
 macro avg          0.87          0.80          0.79       394
 weighted avg       0.85          0.80          0.78       394
```

Figure 16: Classification Report for VGG19

From the classification report analysis, the model demonstrates a solid performance. An accuracy of 80% is achieved whilst a precision of 85% indicates a high proportion of the positive predictions made were correct. Moreover, the model achieves an 80% recall score and a 78% F1-score. Despite these metrics, it is evident that it similarly struggles to detect glioma tumors as only 38% of the instances are correctly classified.

Xception

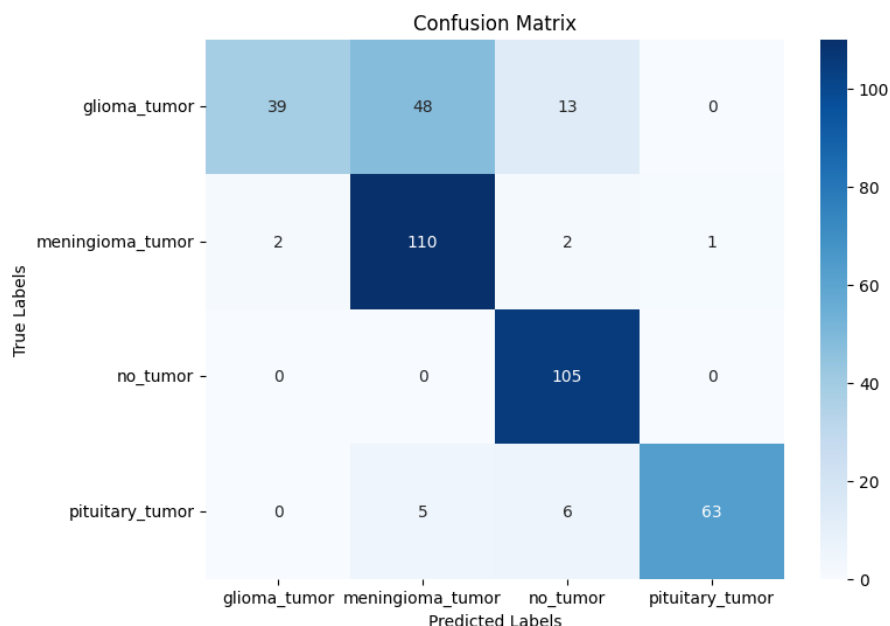


Figure 17: Xception's confusion matrix

The Xception model also finds difficulty distinguishing glioma tumors from meningioma tumors, as 48 samples were misclassified as such. Conversely, the model's superior performance in classifying the meningioma and no tumor classes is exhibited in the confusion matrix. Most pituitary tumor instances, 63 out of 74, were similarly correctly classified.

Classification Report:				
	precision	recall	f1-score	support
glioma_tumor	0.95	0.39	0.55	100
meningioma_tumor	0.67	0.96	0.79	115
no_tumor	0.83	1.00	0.91	105
pituitary_tumor	0.98	0.85	0.91	74
accuracy			0.80	394
macro avg	0.86	0.80	0.79	394
weighted avg	0.85	0.80	0.79	394

Figure 18: Xception's Classification Report

Like the VGG19 model, the Xception model records an accuracy of 80%. It also achieves a precision of 85%, a recall score of 80%, and an F1 score of 79%.

Custom CNN

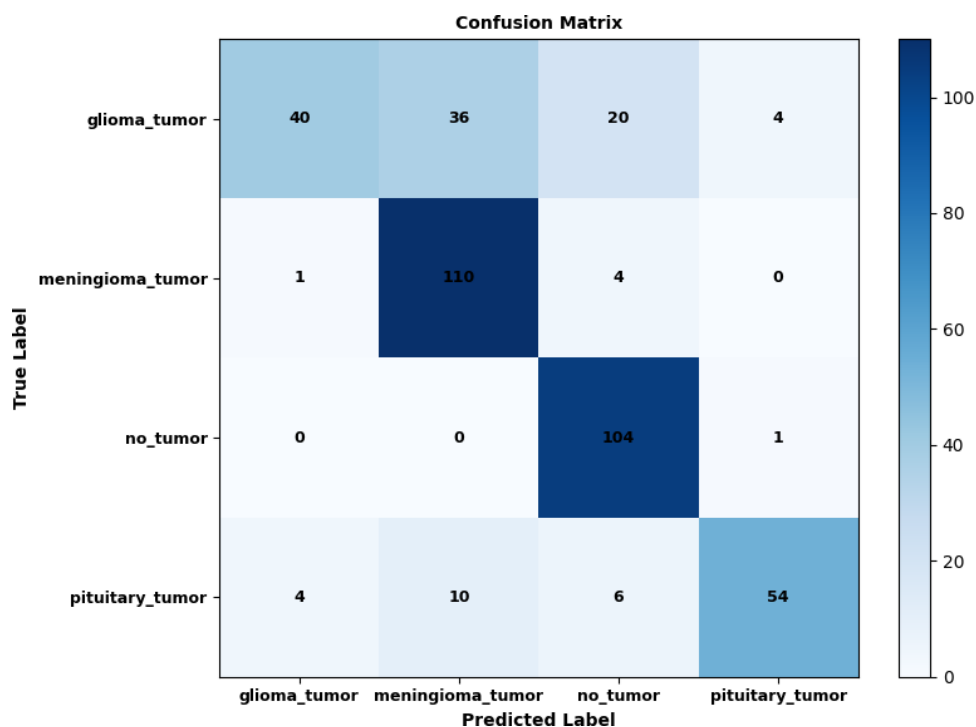
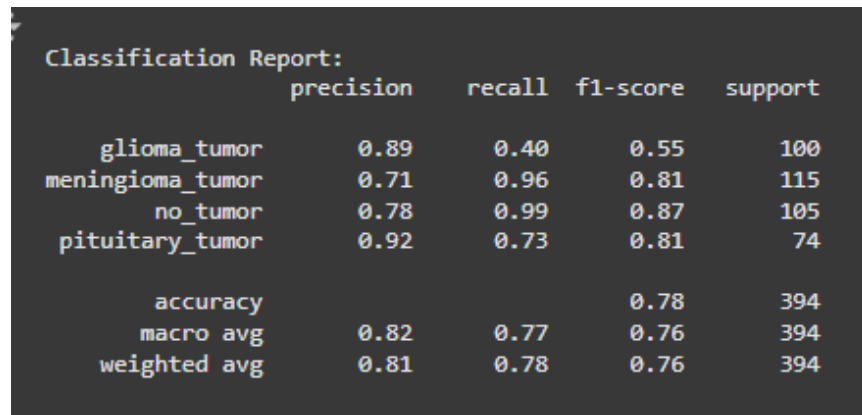


Figure 19: Custom CNN confusion matrix

The custom CNN equally shows good performance in classifying the no-tumor and meningioma tumors. However, like the tested models above, it struggles to classify the glioma tumor with only 40 cases being accurately determined. The model records an overall accuracy of 78%, 78% recall, 81% precision, and 76% F1-score. These results demonstrate that there's room to improve this model through further fine-tuning.



Classification Report:				
	precision	recall	f1-score	support
glioma_tumor	0.89	0.40	0.55	100
meningioma_tumor	0.71	0.96	0.81	115
no_tumor	0.78	0.99	0.87	105
pituitary_tumor	0.92	0.73	0.81	74
accuracy			0.78	394
macro avg	0.82	0.77	0.76	394
weighted avg	0.81	0.78	0.76	394

Figure 20: Custom CNN classification report

Training and Validation Curves

Training and validation curves, specifically the training and validation accuracy curves and the training and validation loss curves are crucial tools in evaluating model training. They are leveraged as visual tools to identify overfitting and to provide insights into which hyperparameters can be adjusted to increase model performance. In these curves, overfitting of the model is depicted by decreasing training loss and an increasing validation loss. Contrastingly, when training and validation loss are high, that indicates underfitting. The following are the curves of the above CNN and transfer learning models.

Transfer Learning Models

The figures below show the training and validation curves for transfer learning models. The accuracy curves show that the models, trained over 40 epochs, were effectively learning, especially after the 10th epoch, when accuracy increased steadily towards 100%. Moreover, the validation accuracy closely tails the training accuracy, indicating minimal overfitting during model training.

Similarly, the plotted loss curves depict that the models converged well as the loss values stabilized at relatively low values. The narrow gap between the loss curves further indicates minimal overfitting. Therefore, this demonstrates the models' effective generalization on unseen data by the transfer learning models.

Xception Curve

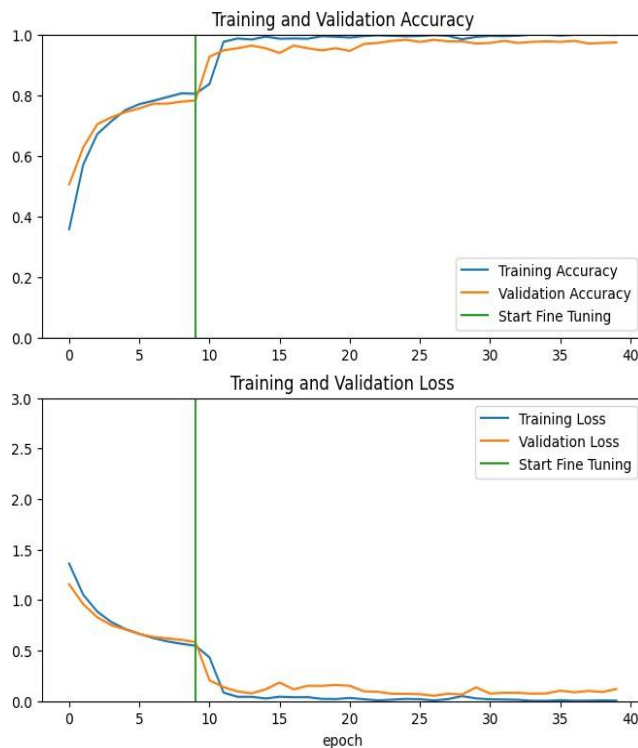


Figure 21: Xception Curve

InceptionV3 Curve

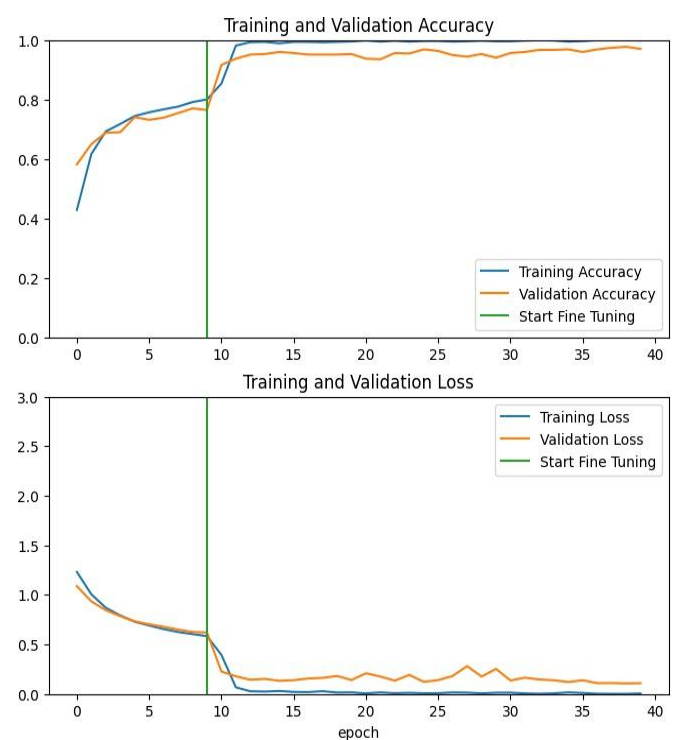


Figure 22: InceptionV3 Curve

VGG19 Curve

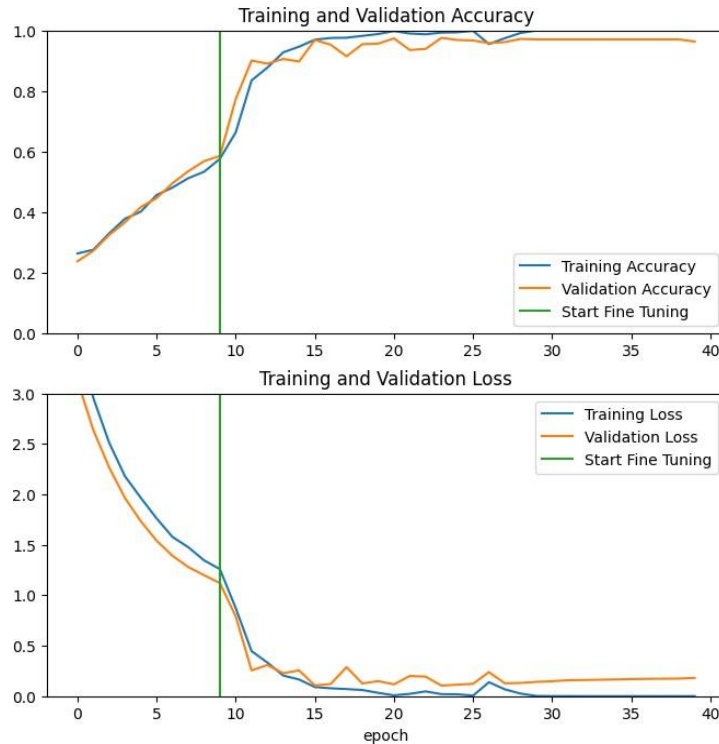


Figure 23: VGG19 Curves

Custom CNN

As mentioned, the custom CNN was experimented over different epochs, with optimal results being achieved over 200 epochs. The curve shows that after the 50th epoch, the model began overfitting. After using early stop callbacks in the initial stages of experimentation, through testing, it proved better to let the models run through the entire epochs. This means that the model was memorizing patterns instead of learning. Consequently, more hyperparameter tuning can be done in future experiments to improve its generalization capacity.

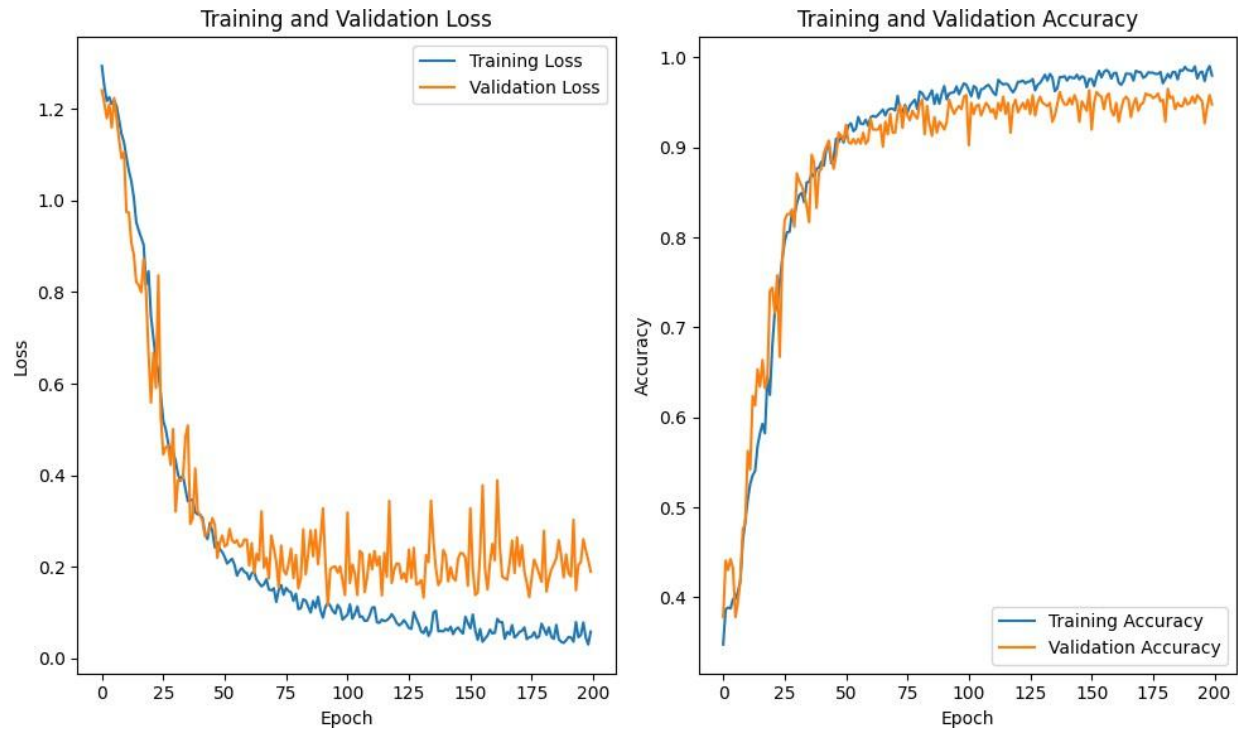


Figure 24: Custom CNN Curves

Table 2: Tabulated Summary of Each Models' Optimal Performance Metrics

Evaluation Metrics				
	Accuracy	Precision	Recall	F1-Score
Model				
<i>Machine Learning</i>				
Logistic Regression	70%	71%	70%	67%
SVM	61%	64%	61%	56%
Decision Tree Classifier	65%	66%	65%	59%
<i>Deep Learning</i>				
Custom - CNN	78%	81%	78%	76%
Xception	80%	85%	80%	79%
VGG19	80%	87%	80%	79%
InceptionV3	86%	89%	86%	85%

From the tabulated results above, it is evident that the deep learning models outperformed the conventional machine learning models in the brain tumor classification task presented in this study. The Custom CNN, which was designed from scratch, indicated good performance, showing the robustness of custom-designed architecture in classifiers.

Nevertheless, the pre-trained CNN models achieved higher performance metrics. The InceptionV3 CNN architecture was the most optimal model, recording an accuracy of 86% and an 86% recall score. Despite this performance, it was evident the model struggled to classify the glioma tumors, as was the case with the rest of the tumors. The overall high recall nevertheless suggests that the model has a high probability of identifying brain tumors in MRI images. Xception and VGG19 had almost similar performances, with the latter edging the former due to its higher precision, 87%.

Additionally, as visualized in the curve below, all models' accuracy and recall values are the same. Therefore, this proves that the models can distinguish between negative and positive instances.

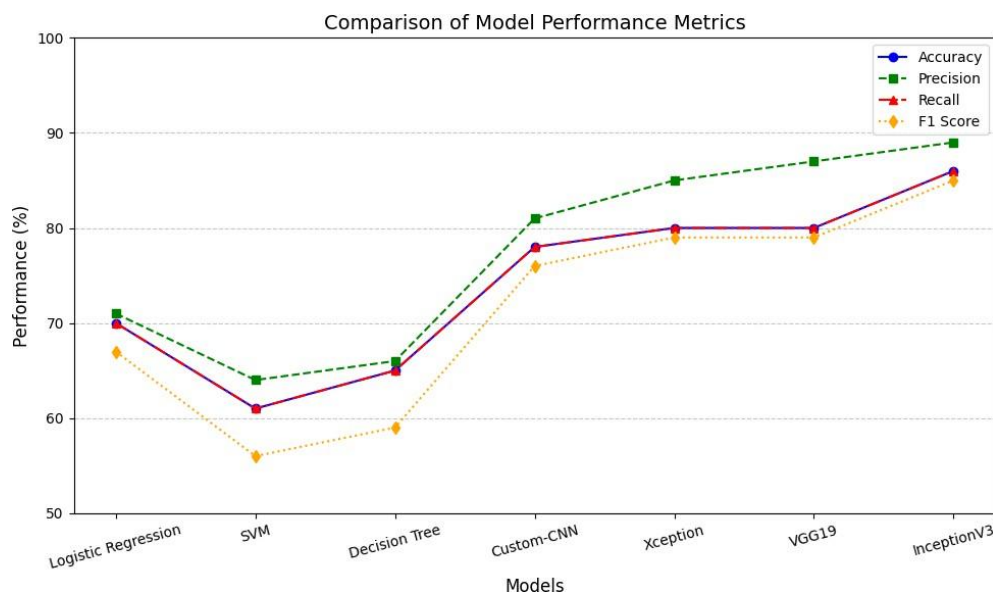


Figure 25: Comparison of Model Performance Metrics

Prediction Results from Custom CNN

The following diagram shows the results posted by the custom convolution neural network in determining the unseen MRI images. The actual class is compared against the predicted class, and the confidence level is similarly determined.

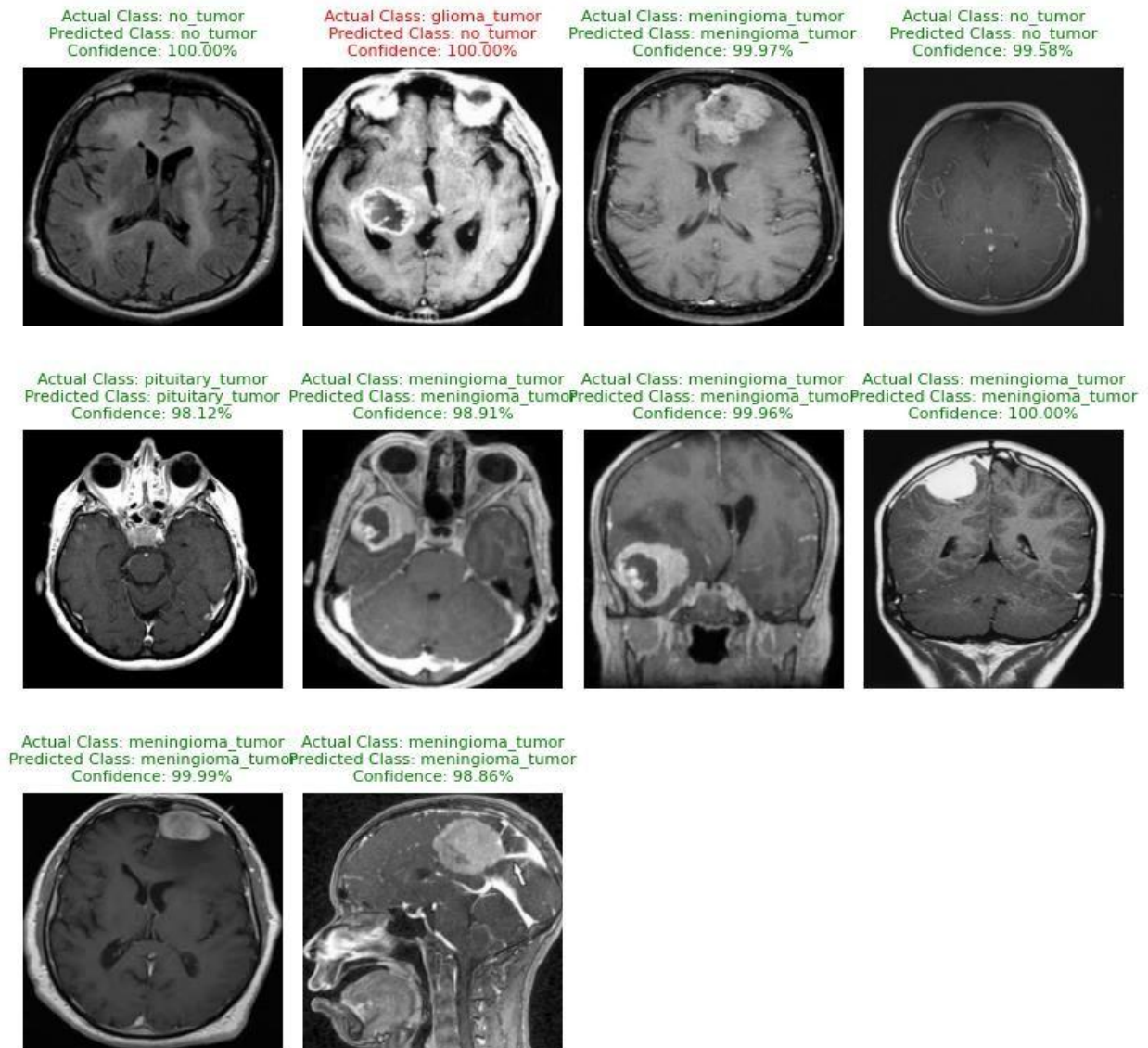


Figure 26: Results of Custom CNN prediction

Chapter 6

6.1 Conclusion

The research delved into brain tumor detection in Kenya with the aim of reducing diagnostic delays. The study followed the CRISP-DM methodology, which provided a structured approach for model development and evaluation. Utilizing a publicly available dataset made a comprehensive analysis comparing machine and deep learning models in brain tumor classification possible.

The findings of this research underwrite the untapped potential of the machine and deep learning technologies in improving brain tumor diagnosis in Kenya. The study evaluated multiple models using the following performance metrics: accuracy, recall, precision, and F1-score. The results indicate that deep learning approaches proved more accurate in this brain tumor detection task. Notably, the pre-trained IncpationV3 was the most optimal model, performing the best, achieving an accuracy of 86% and an 86% recall.

Moreover, the results of this research align with the insights drawn during the literature review. This is exemplified by Table 1 results that compare the performance of the different models described in chapter two, clearly indicating the optimal performance of deep learning models. Consequently, this study corroborates the efficacy of deep learning technologies in brain tumor diagnosis, as demonstrated, for instance, by Abdusalomov et al. (2023).

In conclusion, the research demonstrates the essence of integrating these technologies into the Kenyan healthcare system to facilitate early and accurate brain tumor diagnosis. Early diagnosis would lead to timely interventions in cases where tumors are detected, decreasing mortality rates. Aside from early diagnosis, it was clear that by the detection system correctly classifying the type of brain tumor, misdiagnosis instances can be decreased.

6.2 Future Work

Future work in this domain should address the dataset. A more elaborate dataset with more classes should be utilized for model training and evaluation, as this will improve generalizability. Researchers must also acquire more powerful computational resources that can effectively and exhaustively conduct hyperparameter optimization to determine optimal model parameters.

Additionally, there should be a focus on deploying such a system in clinical workflows to observe and improve them in real-world scenarios. Furthermore, more techniques, such as ensemble learning, can be investigated, and collaborative efforts between researchers should be harnessed.

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